Course Code: ITA1439 **Course Name: Ethical Hacking**

Slot:A

CO2 – ASSIGNMENT_2

Nature of Assignment: Individual **Duration: 1 Week** **Total Marks: 50**

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CO2: Execute footprinting and social-engineering techniques such as DNS analysis, competitive intelligence gathering and attack simulations to collect information for identifying potential security risks.

Assignment Title: Evaluation and Engineering of Footprinting, Reconnaissance and Social-Engineering-Awareness Workflows in an Authorized Assessment

Scenario - Assessment Design Philosophy

The assignment is designed for high-performing cybersecurity students and emphasizes the evaluation of reconnaissance and information-gathering choices rather than the mechanical use of tools. Students are expected to formulate an information-gathering strategy, construct difficult test conditions, run footprinting against authorized targets, collect evidence, identify limitations, compare alternatives and defend engineering decisions. All work is performed only against authorized or lab-owned targets, and no attack is launched against live systems or real individuals.

The assessment follows: Understand → Predict → Design → Collect → Analyse → Evaluate → Defend → Validate.

Q1 – Evaluate and Engineer a Footprinting Strategy **8 Marks**

Problem Context – Multi-Source Open-Source Information Gathering

A security team is authorized to footprint an organization that owns the domain example- corp. example before a penetration test. Each engagement must gather information from WHOIS, DNS, search engines and public documents, organize it, and use it to understand the target's attack surface.

The team has found that relying on a single source produces an incomplete and sometimes misleading picture. A domain with a clean WHOIS record may still leak information through

DNS, and a quiet DNS footprint may still be exposed through public documents. The strategy must therefore combine several sources and judge their reliability rather than trusting one.

Assessment Challenge

Design a footprinting strategy that combines passive and active techniques (WHOIS, DNS records, search-engine operators and archived pages) and classify each technique as passive or active.

Collect information from an authorized domain using appropriate tools such as whois, nslookup and dig, and organize the findings by source.

Ensure that out-of-scope domains, personal data and unverified results are not treated the same as authorized, verified, in-scope findings.

Construct four contrasting targets: a well-protected domain; a domain that leaks information; a domain with misconfigured DNS; and a domain with a minimal public footprint.

For every target, predict what information will be found before collection and compare it with the actual result.

Record the sources queried and the records found, and identify cases where one source contradicts another.

Evaluate whether the chosen sources are reliable and whether any source produces misleading or outdated information.

Changed Condition: The scope grows from one domain to fifty subdomains and several third-party services. Evaluate the original strategy, modify it where necessary, and defend whether it remains suitable for large-scale authorized reconnaissance.

Evidence Expected: Footprinting strategy design, authorized test targets, collected-source records, prediction-versus-observation analysis, source-reliability analysis, scope analysis and a justified final strategy.

Q2 – Build a Reliable DNS Footprinting Workflow **7 Marks**

Problem Context – Conflicting and Misconfigured DNS Data

An authorized assessment must map a target's DNS by retrieving A, MX, NS and TXT records and, only where explicitly permitted, attempting a zone transfer. Records may be inconsistent,

delegated to third parties, cached or misconfigured, so the workflow must interpret the results carefully rather than reporting them at face value.

Assessment Challenge

Develop a DNS footprinting workflow using dig and nslookup that retrieves A, MX, NS and TXT records, and attempts a zone transfer only where authorized.

Evaluate how split-horizon DNS, delegation and caching change the information visible to an external tester.

Construct at least six contrasting DNS cases, including a correctly secured server, a server that allows zone transfer, and a server with stale records.

For each case, predict the records exposed, run the workflow, trace what each query returns, and compare the result with the prediction.

Evaluate whether a misconfigured zone transfer meaningfully changes the attacker's knowledge and the resulting risk.

Introduce a countermeasure such as restricting transfers or using TSIG, and redesign the analysis to show the reduced exposure.

Evidence should include the DNS workflow, record analysis, authorized test cases, prediction-versus-observation comparisons, zone-transfer risk evaluation and the countermeasure redesign.

Evidence Expected: DNS workflow, record analysis, authorized test cases, prediction-versus-observation, zone-transfer risk evaluation and countermeasure redesign.

Q3 – Create a Competitive-Intelligence Awareness Utility **7 Marks**

Problem Context – Public Information and Human-Risk Exposure

A department is building a repeatable workflow that collects public business information about an authorized organization from job advertisements, annual reports, press releases and public profiles, and assesses how that information could support social-engineering attacks such as phishing, vishing, pretexting and tailgating, so that awareness and countermeasures can be designed. No social-engineering attack is performed against real people.

Your Challenge

Design documented steps for each public source and record where each item came from, using search-engine operators such as site, filetype: and intitle.

Map each collected item to the social-engineering technique it could enable (phishing, vishing, pretexting or tailgating) and the indicator a target should notice.

Create test organizations containing rich, sparse and misleading public information.

Predict the information and the human-risk level expected from each source before collection.

Evaluate whether the collected information meaningfully increases social-engineering risk and which source is the most dangerous.

Add two new sources or two new awareness countermeasures without restructuring the whole workflow, and evaluate the effect.

Record the sources queried, the items found and the countermeasures recommended.

Inquiry

Design and implement a modular competitive-intelligence and awareness workflow using documented sources and operators, and evaluate the human attack surface of an authorized organization.

Evidence should include the workflow architecture, source operators, item-to-technique mapping, test organizations, prediction-versus-observation and recommended countermeasures.

Evidence Expected: Workflow architecture, source operators, item-to-technique mapping, test organizations, prediction-versus-observation, risk evaluation and recommended countermeasures.

Q4 – Design a Reconnaissance Data-Validation Mechanism **6 Marks**

Problem Context – High-Stakes Reconnaissance Data Quality

A reconnaissance data-collection process receives results that may be incomplete, outdated, duplicated or out of scope. Acting on poor-quality data wastes effort and can breach the agreed scope, so each item must be validated before it is used.

Your Challenge

Develop a validation mechanism that checks each collected item for scope, freshness, duplication and source reliability.

Distinguish a recoverable item (one that needs a re-query or confirmation) from an item that must be rejected (out of scope or unverifiable).

Construct at least five test sequences, including all-valid, out-of-scope, stale and duplicated data.

Predict how many items pass, fail or need re-checking before running the validation.

Evaluate whether the order of the validation checks affects effort and accuracy.

Redesign the validation order to reduce unnecessary checks while still keeping out-of-scope data out.

Scalability Inquiry

Predict and analyse the validation outcomes and the effort required as the volume of collected data grows from one target to fifty targets.

Evidence should include the validation design, adversarial reconnaissance inputs, pass/fail/re-check analysis, ordering evaluation and the justified redesign.

Evidence Expected: Validation design, adversarial reconnaissance sequences, pass/fail/re-check analysis, ordering evaluation and a justified redesign.

Q5 – Evaluate Footprinting of a Hierarchical Organization **7 Marks**

Problem Context – Irregular Multi-Level Organizational Footprint

A target organization has a nested public footprint of a parent domain, subdomains, departments and third-party services. Footprinting must traverse this hierarchy level by level, recording each level and stopping cleanly at out-of-scope or terminal nodes.

Your Challenge

Represent the organization's public footprint as a hierarchy (domain, subdomains and services) that can be traversed systematically.

Develop a systematic traversal that records each level and stops at out-of-scope or terminal nodes using an explicit boundary condition.

Construct shallow, deeply nested and irregular footprints with branches of differing size.

Predict the number of nodes and the information expected before the traversal.

Evaluate how depth and branching affect the effort and completeness of the footprinting.

Include an empty branch or a node with no public data and evaluate whether the boundary condition handles it correctly.

Compare the systematic traversal conceptually with an ad-hoc, unstructured approach.

Test Your Design

Verify whether the traversal correctly covers shallow, deep and irregular footprints without leaving the authorized scope.

Correctness Inquiry

Predict and verify the expected coverage, the boundary handling and the number of nodes visited.

Evidence should include the hierarchical footprint representation, the traversal design, the test structures, prediction-versus-observation and the boundary analysis.

Evidence Expected: Hierarchical footprint representation, traversal design, test structures, prediction-versus-observation, boundary analysis and a justified final design.

Q6 – Evaluate Footprinting Approaches Under Constraints **7 Marks**

Problem Context – Reconnaissance Under Time and Stealth Constraints

An authorized assessment must gather enough information while balancing speed, completeness and the risk of detection. Passive open-source collection and more active footprinting have different trade-offs, so the better approach depends on the scope and the constraints.

Your Challenge

Select two information-gathering approaches, for example passive open-source collection versus active DNS and service footprinting, and define what each produce.

Apply both approaches to the same authorized target.

Test at least four increasing scope sizes, including one at which one approach becomes impractical.

Record the number of queries or steps and the time taken where useful.

Evaluate redundant or repeated collection in each approach and identify where effort is wasted.

Determine whether the faster or the quieter approach is always the better engineering choice.

Defend a recommendation for a small-scope engagement and for a large-scope engagement.

Inquiry

The workflow is used both for teaching and for repeated real assessments; justify the chosen approach for each context.

Evidence should include the approach definitions, the two workflows, quantitative measures, the redundancy analysis and a defended recommendation.

Evidence Expected: Approach definitions, two workflows, quantitative measures, redundancy analysis and a defended recommendation.

DAY-4 DESIGN CHALLENGE

Q7 – Integrated Review: Diagnose, Redesign and Defend **8 Marks**

Select one workflow from Q1–Q6 that demonstrates a meaningful limitation, such as incomplete coverage, an unreliable source, excessive queries, weak validation or a risk of leaving scope, and improve it using evidence.

A. Diagnose the Limitation **2 Marks**

Identify the reconnaissance or logical bottleneck using evidence.

Determine whether the limitation is caused by source unreliability, redundant collection, poor ordering or a risk of going out of scope.

EVIDENCE EXPECTED

Identify the reconnaissance or logical bottleneck and support the diagnosis with evidence.

Pinpoint the underlying cause of the limitation, such as redundant collection, an unreliable source or poor ordering.

B. Establish Evaluation Criteria **1 Mark**

Define measurable criteria against which the original workflow will be judged.

Examples include coverage, the number of queries, source reliability, scope compliance and time.

Evidence Expected

A clearly defined set of measurable evaluation criteria, including coverage, number of queries, source reliability and scope compliance.

C. Propose and Implement a Redesign **2 Marks**

Modify the original workflow to address the diagnosed limitation.

Provide the design logic or the ordered steps before implementation.

Identify any trade-off introduced by the redesign.

Evidence Expected

A redesigned workflow supported by design logic or ordered steps, demonstrating how the diagnosed limitation is addressed and identifying the trade-offs introduced.

D. Validate and Compare **2 Marks**

Use equivalent test conditions for the original and the redesigned workflow.

Collect quantitative evidence and determine whether the redesign actually improves the selected criterion.

Evidence Expected

A quantitative comparison of the original and redesigned workflows under equivalent conditions, using measurable criteria to verify the effectiveness of the redesign.

E. Defend the Final Engineering Judgment **1 Mark**

Evaluate whether the redesign met its objective.

Explain cases in which the original workflow may still be preferable.

Defend the final choice using experimental evidence rather than subjective statements.

Evidence Expected

A justified final engineering decision supported by experimental results, evaluating the redesign's effectiveness and identifying cases in which the original workflow remains preferable.

MARK DISTRIBUTION

Question	Assessed Area	Marks
Q1	Footprinting strategy design and evaluation	8
Q2	DNS footprinting and analysis workflow	7
Q3	Competitive-intelligence and awareness utility	7
Q4	Reconnaissance data validation and recovery	6
Q5	Hierarchical organization footprinting	7
Q6	Footprinting approach comparison	7
Q7	Integrated diagnose–redesign–validate challenge	8
TOTAL		**50**

Submission Requirements

Submit:

Assignment report containing evidence for Q1–Q7.

Scripts or documented command sequences for all implemented workflows.

Student-designed authorized test targets and adversarial cases.

Raw collected data and tool outputs.

Prediction-versus-observation comparisons.

Tables/graphs wherever they strengthen the evaluation.

Source, query, item and scope counts where applicable.

Design representation or pseudocode for redesigned workflows.

Execution time may be reported as supporting evidence, but conclusions should not rely solely on timing; coverage, source reliability and scope compliance should form the primary evidence.

Assessment Verification / Viva

Students may be asked to explain and defend their source selection, footprinting methodology, DNS and open-source analysis, scope and authorization decisions, social-engineering-awareness reasoning, adversarial test design, measured results, redesign choices and final engineering judgments.

Overall Assessment Expectation

The assignment assesses the student's ability to evaluate information-gathering solutions under realistic, authorized conditions: Problem Interpretation → Prediction → Design → Collection → Adversarial Testing → Measurement → Analysis → Evaluation → Defence → Engineering Judgment.

1. Evaluate and Engineer a Footprinting Strategy

1. Aim

To design a reliable footprinting strategy that combines multiple information sources, identifies the public attack surface, and prevents out-of-scope or unverified information from being treated as confirmed findings.

2. Footprinting strategy

Footprinting is the process of collecting information about an organization before a security assessment.

The strategy uses the following sources:

Technique	Type	Information collected
WHOIS	Passive	Domain registration and registrar details
Search engines	Passive	Public pages, documents, and subdomains
Archived pages	Passive	Previously published information
DNS queries	Passive when querying public resolvers	A, MX, NS, and TXT records
Direct DNS queries to an authorized server	Active	DNS information visible through direct interaction

3. Workflow

Start
|
v
Confirm authorization
|
v
Define assessment scope
|

v

Collect WHOIS information

|

v

Collect public DNS records

|

v

Search public pages and documents

|

v

Check archived information

|

v

Validate source and scope

|

v

Compare and classify findings

|

v

Prepare final report

4. Authorized test targets

Four fictional test domains are used to evaluate the strategy.

Target	Description	Predicted information
Target A	Well-protected domain	Limited public information and correctly configured DNS
Target B	Information-leaking domain	Public documents and additional subdomains
Target C	Misconfigured DNS domain	Inconsistent or outdated DNS records
Target D	Minimal-footprint domain	Very little publicly available information

5. Prediction versus observation

Target	Prediction	Sample observation
A	Limited information	Only basic registration and DNS information found
B	Public documents and subdomains	Three public documents and two subdomains found
C	DNS inconsistencies	One stale DNS record and an outdated archived page found
D	Minimal information	Only basic domain and nameserver information found

The observations are consistent with the predictions. However, a limited public footprint does not prove that a domain is secure.

6. Source reliability

- WHOIS records may be outdated or privacy-protected.
- DNS records may be cached and may not reflect current configuration.
- Search-engine results may contain old or removed pages.
- Archived pages may contain information that is no longer valid.
- Public documents may have been published without proper review.

Therefore, important findings should be confirmed using a second reliable source.

7. Changed condition: Fifty subdomains and third-party services

The original strategy must be modified because manually checking every subdomain and service becomes time-consuming.

The improved strategy includes:

1. Maintaining an approved scope list.
2. Automating permitted information collection.
3. Organizing results in a structured database or spreadsheet.
4. Removing duplicate findings.
5. Validating the source and freshness of every finding.
6. Separating third-party services from directly owned infrastructure.
7. Reviewing results before including them in the final report.

Final evaluation: The improved strategy is suitable for large-scale authorized reconnaissance because it increases consistency, reduces duplication, and maintains scope control. However, automated collection still requires human review.

Q2. Build a Reliable DNS Footprinting Workflow

1. Aim

To collect and analyze DNS records accurately while identifying misconfigurations, stale records, and unauthorized zone-transfer exposure.

2. DNS workflow

Start

|

v

Confirm authorized domain

|

v

Query A, MX, NS and TXT records

|

v

Record DNS responses and TTL

|

v

Compare with expected configuration

|

v

Check for stale or conflicting records

|

v

Attempt zone transfer only if authorized

|

v

Apply security countermeasures

|

v

Prepare report

3. DNS commands

The following commands are examples for an authorized lab domain.

dig example.test A
dig example.test MX
dig example.test NS
dig example.test TXT

Equivalent nslookup commands:

nslookup -type=A example.test
nslookup -type=MX example.test
nslookup -type=NS example.test
nslookup -type=TXT example.test

An authorized zone-transfer test can be performed in a lab using:

dig @ns1.example.test example.test AXFR

A successful transfer should be reported only when it is confirmed that the server is improperly allowing transfers to unauthorized clients.

4. Six DNS test cases

Case	Configuration	Predicted result	Sample observation
1	Correctly secured DNS	Valid records; unauthorized transfer denied	Prediction confirmed
2	Zone transfer enabled for unauthorized clients	Excessive DNS information exposed	Transfer succeeded in lab
3	Stale A record	Old IP address returned	Old address observed
4	Split-horizon DNS	Different internal and external responses	External response differed
5	Third-party DNS delegation	Records controlled by external provider	Delegated nameservers observed
6	Cached DNS response	Older record may appear temporarily	Cached value differed from authoritative value

5. Zone-transfer risk evaluation

An unauthorized zone transfer may reveal numerous subdomains and infrastructure names.

This increases the amount of information available to an attacker and can make further reconnaissance easier.

However, a successful transfer does not automatically mean the organization has been compromised.

6. Countermeasure redesign

- Restrict zone transfers to approved secondary DNS servers.
- Use TSIG to authenticate authorized transfer requests.
- Remove stale DNS records.
- Review third-party DNS delegation.
- Monitor DNS configuration changes.

Conclusion: The redesigned DNS workflow reduces unnecessary information exposure and improves the accuracy of reconnaissance results.

Q3. Create a Competitive-Intelligence Awareness Utility (7 Marks)

1. Aim

To collect public business information and understand how it could increase social-engineering risk, without attacking real people.

2. Workflow architecture

Authorized Organization

|

v

Collect Public Information

|

v

+----------------------------+

| Job Advertisements |

| Annual Reports |

| Press Releases |

| Public Business Profiles |

+--------------------------+

|

v

Record Source and Date

|

v

Identify Human-Risk Exposure

|

v

Map to Social-Engineering Risks

|

v

Recommend Awareness Measures

|

v

Prepare Report

3. Search operators

Operator	**Purpose**
site:	Search within a specific website
filetype:pdf	Find public PDF documents
intitle:	Search pages with a particular title

Examples:

site:example.test careers

site:example.test filetype:pdf annual report

site:example.test intitle:press release

These searches should only be used within the approved assessment scope.

4. Test organizations

Organization	**Public information**	**Predicted risk**
RichInfo	Detailed job advertisements, reports, and press releases	High
SparseInfo	Basic website and limited public information	Low

MisleadInfo	Outdated or incorrect public information	Medium

5. Item-to-technique mapping

Public information	Possible technique	Warning indicator	Countermeasure
Job advertisements	Phishing	Unexpected recruitment links	Verify the sender
Annual reports	Pretexting	Requests using internal project details	Verify identity independently
Public contact numbers	Vishing	Urgent requests for confidential information	Call back using official numbers
Office location information	Tailgating	Unknown person requesting entry	Follow visitor procedures

6. Prediction versus observation

Source	Prediction	Sample observation
Job advertisements	Reveal departments and technologies	Several department names found
Annual reports	Reveal business operations	Business structure identified
Press releases	Reveal projects and partnerships	Two public projects identified
Public profiles	Reveal employee roles	Several public job titles found

The results show that public business information can increase the credibility of social-engineering attempts.

7. Additional countermeasures

Two additional awareness measures are recommended:

1. Conduct regular phishing-awareness training using fictional messages.
2. Introduce a verification procedure for requests involving confidential information.

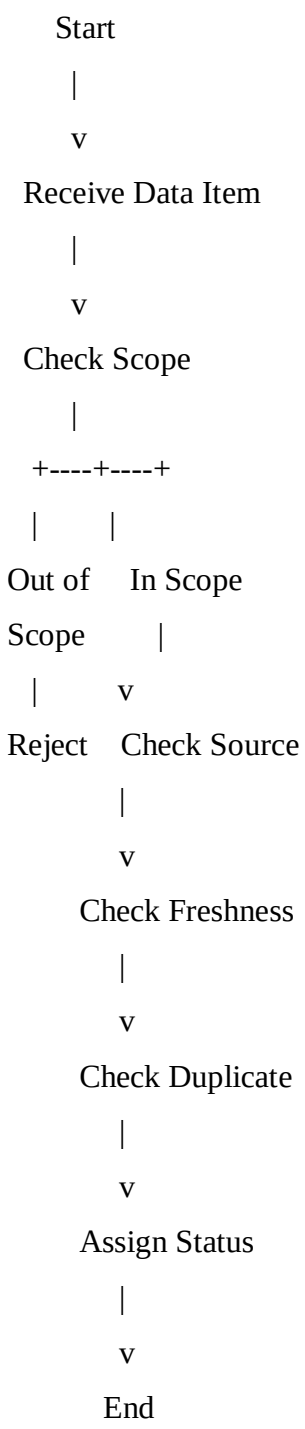
Conclusion: Public information is useful for business research, but it should be reviewed for unnecessary exposure. Employee awareness and verification procedures reduce the associated human risk.

Q4. Design a Reconnaissance Data-Validation Mechanism

1. Aim

To ensure that collected reconnaissance data is in scope, current, unique, and supported by reliable sources.

2. Validation workflow



3. Validation rules

Check	Condition	Action

Scope	Item is outside approved scope	Reject
Source	Source cannot be verified	Reject
Freshness	Information is outdated	Re-check
Duplicate	Same finding already exists	Merge
All checks	Item is valid	Accept

4. Test sequences

	Test Input	**Predicted outcome**
1	Valid, fresh, in-scope item	Pass
2	Out-of-scope domain	Reject
3	Stale DNS record	Re-check
4	Duplicate public document	Merge
5	Unverifiable information	Reject

5. Sample validation results

Assume the following seven items are collected.

Item	**Status**	**Reason**
1	Pass	Valid and current
2	Pass	Valid and current
3	Reject	Out of scope
4	Re-check	Stale record
5	Merge	Duplicate
6	Reject	Unverifiable
7	Pass	Valid and current

Summary:

- Passed: 3
- Rejected: 2
- Re-check: 1
- Duplicate merged: 1

6. Improved validation order

The improved order is:

1. Check scope first.
2. Check whether the source is reliable.
3. Check for duplication.
4. Check freshness.
5. Accept, reject, merge, or re-check the item.

Checking scope first prevents unnecessary work on unauthorized data.

Conclusion: The validation mechanism improves data quality and reduces the risk of acting on outdated or out-of-scope information.

Q5. Evaluate Footprinting of a Hierarchical Organization

1. Aim

To represent an organization's public footprint as a hierarchy and traverse it systematically without crossing the authorized boundary.

2. Hierarchical representation

Organization Domain

```
|
+-- Department A
|    |
|    +-- Service A1
|    +-- Service A2
|
+-- Department B
|    |
|    +-- Service B1
|
+-- Department C
     |
     +-- No Public Data
```

Each node represents a domain, subdomain, department, or public service.

3. Traversal algorithm

Start at the authorized root

|

v

Record current node

|

v

Is node in scope?

| |

No Yes

| |

Stop Check children

|

v

Are children present?

| |

No Yes

| |

Stop Visit each child

|

v

Continue

4. Test structures

Structure	Description	Expected nodes
Shallow	Root with three children	4
Deep	Root with a chain of five descendants	6
Irregular	Root with branches of different sizes	7

5. Prediction versus observation

Test	Predicted nodes	Sample visited nodes	Coverage
Shallow	4	4	100%
Deep	6	6	100%

Irregular	7	7	100%

Coverage is calculated as:

$$\text{Coverage}= \frac{\text{Authorized nodes visited}} {\text{Total authorized nodes}}\times100$$

6. Boundary analysis

The traversal must stop when:

- A node is outside the approved scope.
- A node has no children.
- A node has no public information.
- A terminal service has been reached.

An empty branch should be reco

rded as having no public data, rather than treated as a traversal failure.

7. Systematic versus ad-hoc traversal

Systematic traversal provides better coverage, traceability, and scope control. Ad-hoc traversal may be faster for a very small target but can miss branches or revisit the same nodes.

Conclusion: A hierarchical traversal is more reliable for large and irregular organizations because it provides predictable coverage and clear stopping conditions.

Q6. Evaluate Footprinting Approaches Under Constraints

1. Aim

To compare passive open-source collection and active DNS footprinting based on time, query count, completeness, and scope.

2. Selected approaches

Approach A – Passive reconnaissance

Uses search engines, public documents, and archived pages.

Advantages:

- Lower direct interaction with target systems.
- Suitable for initial information gathering.
- Usually produces fewer direct network queries.

Limitations:

- Information may be incomplete or outdated.
- Some subdomains may not appear in public search results.

Approach B – Active DNS footprinting

Uses authorized DNS queries to identify records and verify DNS information.

Advantages:

- Provides current DNS responses.
- Helps identify configuration inconsistencies.

Limitations:

- Generates direct queries.
- Requires authorization and careful rate control.

3. Experimental test conditions

The following sample measurements are illustrative.

Scope size	Passive queries	Active DNS queries	Passive time	Active time
1 target	5	8	2 min	1 min
5 targets	20	40	5 min	4 min
20 targets	60	160	12 min	15 min
50 targets	150	400	30 min	45 min

4. Redundancy analysis

Passive collection may retrieve the same document through multiple search engines.

Active DNS collection may repeat queries for records that have already been checked.

Both approaches should use a deduplication mechanism and a record of previously collected results.

5. Engineering recommendation

For small-scope engagements, passive reconnaissance should be performed first, followed by targeted active DNS verification.

For large-scope engagements, automated collection with scope validation, deduplication, and rate limits is more suitable.

Conclusion: The fastest approach is not always the best. The preferred method depends on the required completeness, authorized scope, available time, and risk of detection.

Q7. Integrated Review: Diagnose, Redesign and Defend (8 Marks)

Selected workflow

Q4 – Reconnaissance Data-Validation Mechanism.

A. Diagnose the limitation (2 Marks)

The original validation process checks source reliability and freshness before checking whether the item is in scope.

This creates unnecessary work because out-of-scope items may undergo several checks before being rejected.

The main limitation is poor validation ordering.

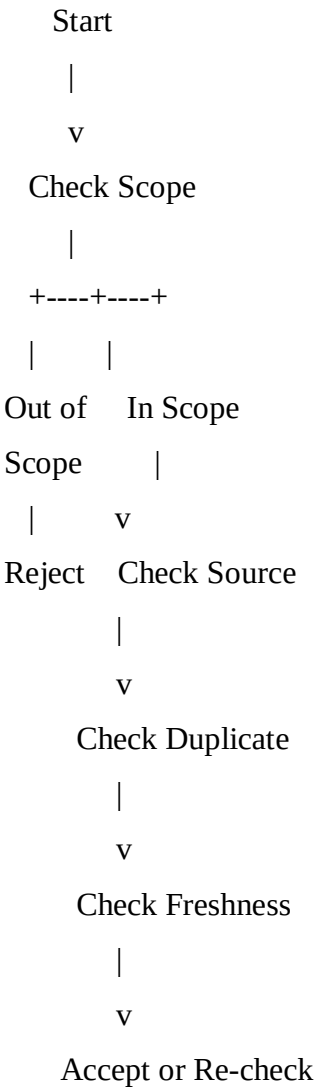
B. Establish evaluation criteria (1 Mark)

The redesign will be evaluated using:

Criterion	**Measurement**
Scope compliance	Number of out-of-scope items accepted
Validation effort	Number of checks performed
Accuracy	Correct classification of items
Completeness	Number of valid items accepted

C. Propose and implement a redesign (2 Marks)

The redesigned workflow checks scope first.



The redesign reduces unnecessary checks while preserving scope control.